

## MIRNY, Russian Federation

WMO: 247260

Lat: 62.535N

Lon: 114.039E

Elev: 1155

StdP: 14.09

Time Zone: 9.00 (E09)

Period: 98-19

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |       | Humidification DP/MCDB and HR |     |       |       |     |       | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|-------|-------------------------------|-----|-------|-------|-----|-------|-----------------------|------|------|------|-----------------------|------|-------|
|               |            |       | 99.6%                         |     |       | 99%   |     |       | 0.4%                  |      | 1%   |      |                       |      |       |
|               | 99.6%      | 99%   | DP                            | HR  | MCDB  | DP    | HR  | MCDB  | WS                    | MCDB | WS   | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)   | (d)                           | (e) | (f)   | (g)   | (h) | (i)   | (j)                   | (k)  | (l)  | (m)  | (n)                   | (o)  | (p)   |
| 12            | -49.0      | -44.8 | -54.0                         | 0.2 | -48.8 | -49.2 | 0.3 | -44.5 | 19.7                  | -0.6 | 18.0 | -4.8 | 2.3                   | 240  | 0.766 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 7             | 18.4                   | 84.1            | 64.5 | 79.6 | 62.6 | 75.6 | 61.1 | 67.1                | 77.9 | 64.9 | 75.8 | 62.8 | 72.9 | 8.2                  | 190  |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 63.0                            | 90.0 | 70.1 | 61.0 | 83.6 | 68.7 | 59.0 | 77.7 | 66.6 | 32.1          | 78.3 | 30.4 | 75.1 | 28.9 | 72.5 | 76.5              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |      | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |      |            |      |            |      |            |      |    |    |
|-------------------|------|------|----------------------------|------|--------------------|-----|----------------------------------------------------|------|------------|------|------------|------|------------|------|----|----|
|                   |      |      | Mean                       |      | Standard Deviation |     | n=5 years                                          |      | n=10 years |      | n=20 years |      | n=50 years |      |    |    |
| 1%                | 2.5% | 5%   | Min                        | Max  | Min                | Max | Min                                                | Max  | Min        | Max  | Min        | Max  | Min        | Max  | DB | WB |
| (a)               | (b)  | (c)  | (e)                        | (f)  | (g)                | (h) | (i)                                                | (j)  | (k)        | (l)  | (m)        | (n)  | (o)        | (p)  |    |    |
| (4) 19.0          | 16.7 | 14.4 | -50.6                      | 90.1 | 5.9                | 2.9 | -54.9                                              | 92.1 | -58.3      | 93.9 | -61.6      | 95.5 | -65.9      | 97.6 |    |    |
| (5)               |      |      | -50.7                      | 70.6 | 5.7                | 2.7 | -54.8                                              | 72.6 | -58.1      | 74.2 | -61.3      | 75.7 | -65.5      | 77.7 |    |    |

## Monthly Climatic Design Conditions

|                                                                        |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |  |  |
|------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|--|--|
|                                                                        |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |  |  |
| (6) Temperatures, Degree-Days and Degree-Hours                         | DBAvg       | 20.7   | -21.1 | -16.1 | 5.2   | 24.4  | 41.6  | 59.0  | 63.2  | 56.6  | 41.2  | 20.4  | -6.7  | -21.7 |  |  |
|                                                                        | DBStd       | 32.75  | 13.80 | 13.33 | 13.40 | 10.14 | 8.70  | 8.69  | 7.03  | 6.62  | 8.00  | 11.36 | 15.30 | 15.00 |  |  |
|                                                                        | HDD50       | 11658  | 2203  | 1851  | 1389  | 767   | 286   | 21    | 2     | 19    | 279   | 917   | 1700  | 2224  |  |  |
|                                                                        | HDD65       | 16273  | 2668  | 2271  | 1854  | 1217  | 726   | 216   | 117   | 270   | 713   | 1382  | 2150  | 2689  |  |  |
|                                                                        | CDD50       | 970    | 0     | 0     | 0     | 0     | 25    | 292   | 413   | 223   | 17    | 0     | 0     | 0     |  |  |
|                                                                        | CDD65       | 108    | 0     | 0     | 0     | 0     | 0     | 37    | 62    | 9     | 0     | 0     | 0     | 0     |  |  |
|                                                                        | CDH74       | 1220   | 0     | 0     | 0     | 0     | 9     | 451   | 640   | 118   | 2     | 0     | 0     | 0     |  |  |
|                                                                        | CDH80       | 330    | 0     | 0     | 0     | 0     | 1     | 118   | 189   | 22    | 0     | 0     | 0     | 0     |  |  |
|                                                                        | WSAvg       | 7.1    | 6.1   | 6.5   | 6.9   | 7.8   | 8.5   | 7.6   | 7.1   | 6.9   | 7.3   | 7.7   | 6.3   | 5.9   |  |  |
|                                                                        | PrecAvg     | 14.9   | 0.6   | 0.5   | 0.6   | 0.8   | 1.3   | 2.0   | 2.0   | 2.2   | 1.5   | 1.6   | 1.1   | 0.8   |  |  |
| (15) Precipitation                                                     | PrecMax     | 20.1   | 1.2   | 1.1   | 1.1   | 1.7   | 2.1   | 3.7   | 3.7   | 3.6   | 2.8   | 2.7   | 1.9   | 3.7   |  |  |
|                                                                        | PrecMin     | 11.2   | 0.2   | 0.2   | 0.1   | 0.2   | 0.6   | 0.7   | 0.7   | 0.7   | 0.2   | 0.5   | 0.4   | 0.2   |  |  |
|                                                                        | PrecStd     | 2.0    | 0.2   | 0.2   | 0.2   | 0.4   | 0.4   | 0.7   | 0.7   | 0.8   | 0.7   | 0.6   | 0.4   | 0.6   |  |  |
|                                                                        |             |        |       |       |       |       |       |       |       |       |       |       |       |       |  |  |
| (19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures | 0.4%        | DB     | 16.2  | 22.7  | 41.1  | 53.6  | 73.3  | 88.0  | 89.6  | 83.4  | 70.2  | 46.8  | 31.9  | 22.9  |  |  |
|                                                                        |             | MCWB   | 15.8  | 21.7  | 33.1  | 41.3  | 54.3  | 65.7  | 67.9  | 63.6  | 55.1  | 40.6  | 31.3  | 22.2  |  |  |
|                                                                        | 2%          | DB     | 8.6   | 15.2  | 34.0  | 48.1  | 66.7  | 83.8  | 85.7  | 76.9  | 62.7  | 41.2  | 28.4  | 14.0  |  |  |
|                                                                        |             | MCWB   | 8.2   | 14.5  | 29.2  | 38.0  | 50.5  | 63.2  | 66.3  | 60.8  | 50.8  | 37.5  | 27.5  | 14.0  |  |  |
|                                                                        | 5%          | DB     | 3.3   | 10.0  | 28.6  | 43.2  | 61.2  | 79.2  | 80.7  | 72.5  | 57.5  | 37.4  | 24.1  | 6.5   |  |  |
|                                                                        |             | MCWB   | 3.1   | 9.2   | 25.6  | 34.8  | 48.2  | 61.4  | 64.2  | 59.0  | 48.8  | 35.2  | 23.5  | 6.4   |  |  |
|                                                                        | 10%         | DB     | -2.0  | 4.0   | 24.7  | 39.3  | 57.0  | 75.3  | 76.9  | 68.4  | 53.5  | 34.0  | 16.1  | -0.2  |  |  |
|                                                                        |             | MCWB   | -2.2  | 3.6   | 21.9  | 32.7  | 45.8  | 59.5  | 62.5  | 57.5  | 46.9  | 32.4  | 15.7  | -0.4  |  |  |
|                                                                        | 0.4%        | WB     | 15.8  | 21.7  | 34.2  | 41.7  | 56.1  | 68.8  | 71.1  | 66.3  | 56.4  | 41.4  | 30.9  | 21.5  |  |  |
|                                                                        |             | MCDB   | 16.8  | 22.7  | 39.2  | 51.3  | 68.8  | 80.1  | 82.6  | 77.0  | 65.7  | 44.6  | 31.0  | 22.0  |  |  |
| (27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures | 2%          | WB     | 8.1   | 14.2  | 29.2  | 38.4  | 52.1  | 65.4  | 68.1  | 63.2  | 53.1  | 38.2  | 27.5  | 12.9  |  |  |
|                                                                        |             | MCDB   | 8.4   | 14.8  | 33.1  | 46.4  | 63.8  | 78.0  | 78.7  | 72.2  | 59.6  | 40.6  | 28.1  | 12.9  |  |  |
|                                                                        | 5%          | WB     | 3.0   | 8.8   | 25.7  | 35.9  | 49.1  | 63.2  | 66.2  | 61.2  | 50.1  | 35.5  | 23.2  | 5.5   |  |  |
|                                                                        |             | MCDB   | 3.2   | 9.6   | 28.7  | 42.0  | 60.5  | 76.2  | 77.0  | 69.0  | 55.5  | 37.2  | 23.8  | 5.6   |  |  |
|                                                                        | 10%         | WB     | -2.3  | 3.4   | 22.0  | 33.4  | 46.3  | 61.1  | 64.4  | 59.2  | 47.9  | 32.6  | 15.5  | -0.8  |  |  |
|                                                                        |             | MCDB   | -2.2  | 3.8   | 24.4  | 38.7  | 55.7  | 72.2  | 74.5  | 66.4  | 53.0  | 33.8  | 16.0  | -0.6  |  |  |
| (35) Mean Daily Temperature Range                                      | 5% DB       | MDBR   | 11.1  | 14.3  | 19.0  | 18.4  | 17.3  | 19.6  | 18.4  | 17.8  | 14.4  | 11.3  | 12.0  | 11.2  |  |  |
|                                                                        |             | MCDBR  | 15.2  | 14.6  | 19.2  | 20.4  | 24.8  | 25.6  | 24.9  | 24.3  | 21.8  | 13.0  | 15.5  | 15.7  |  |  |
|                                                                        |             | MCWBR  | 15.1  | 13.8  | 15.6  | 13.5  | 14.0  | 11.2  | 10.6  | 11.4  | 12.5  | 9.9   | 15.6  | 16.0  |  |  |
|                                                                        |             | MCDBR  | 15.2  | 14.5  | 18.4  | 17.9  | 23.1  | 22.5  | 21.3  | 20.3  | 18.7  | 12.0  | 15.4  | 16.3  |  |  |
|                                                                        | 5% WB       | MCWBR  | 15.1  | 13.9  | 15.1  | 12.7  | 14.2  | 11.5  | 11.4  | 11.3  | 12.3  | 10.0  | 15.6  | 16.2  |  |  |
|                                                                        |             |        |       |       |       |       |       |       |       |       |       |       |       |       |  |  |
| (40) Clear-Sky Solar Irradiance                                        | taub        |        | 0.219 | 0.261 | 0.293 | 0.326 | 0.354 | 0.399 | 0.435 | 0.422 | 0.320 | 0.276 | 0.213 | 0.190 |  |  |
|                                                                        | taud        |        | 1.929 | 2.053 | 2.080 | 2.124 | 2.318 | 2.310 | 2.195 | 2.180 | 2.477 | 2.316 | 1.962 | 1.873 |  |  |
|                                                                        | Ebn at Noon |        | 132   | 215   | 254   | 270   | 273   | 261   | 245   | 236   | 251   | 214   | 132   | 79    |  |  |
|                                                                        | Edh at Noon |        | 12    | 23    | 32    | 39    | 35    | 36    | 40    | 37    | 23    | 17    | 11    | 6     |  |  |
| (44) All-Sky Solar Radiation                                           | RadAvg      |        | 97    | 392   | 935   | 1544  | 1727  | 1821  | 1650  | 1265  | 764   | 358   | 130   | 44    |  |  |
|                                                                        | RadStd      |        | 5     | 23    | 39    | 77    | 108   | 118   | 112   | 75    | 77    | 30    | 12    | 2     |  |  |

## Historical Trends

|      |                        | DBAvg | Heating |        |       | Cooling |       |     | Degree-Days |       |       |       |
|------|------------------------|-------|---------|--------|-------|---------|-------|-----|-------------|-------|-------|-------|
|      |                        |       | 99% DB  | 99% DP | 1% DB | 1% WB   | 1% DP |     | HDD50       | HDD65 | CDD50 | CDD65 |
|      |                        | (d)   | (e)     | (f)    | (g)   | (h)     | (i)   | (j) | (k)         | (l)   | (m)   |       |
| (46) | Station Only           | N/A   | N/A     | N/A    | N/A   | N/A     | N/A   | N/A | N/A         | N/A   | N/A   |       |
| (47) | Regional (3 neighbors) | +1.37 | N/A     | N/A    | N/A   | N/A     | N/A   | N/A | N/A         | N/A   | N/A   |       |

Nomenclature: See separate page